

SurgVU Challenge 2026 — Team **APC_TDC**

Team Report

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0.1 APC_TDC

Code Repository: <https://github.com/seantangth/miccai2026-surgvu-surgical-tool-detection-vqa>

We entered both categories. Our Category 1 (C1) result rests on one finding from controlled comparisons: the dominant lever here is the scale and scale-diversity of pseudo-labelled training data, not the architecture or the recipe. Every recipe change we tried moved the official score by at most 0.006; growing the pseudo-label source from 20 to 280 videos moved it by +0.052. We therefore spent our compute on an iterative pseudo-labelling loop that uses the tool-presence labels as an external gate on a teacher ensemble. For Category 2 (C2) we answer with a rule-based router rather than a generative model.

0.1.1 Method Description

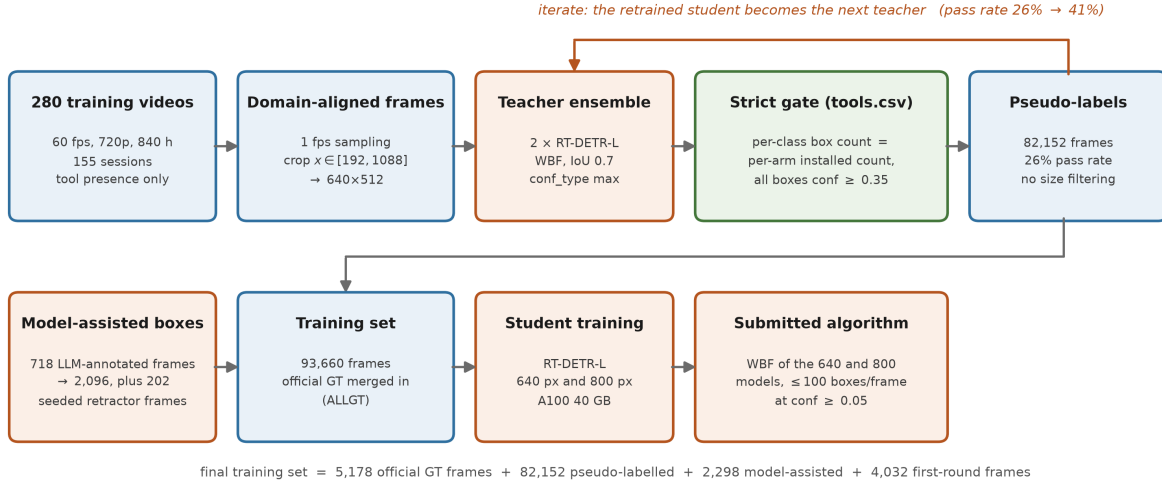


Figure 1: Category 1 pipeline. Tool-presence labels act as an external gate on a teacher ensemble; the accepted frames train a student that becomes the next teacher.

Domain alignment. Public validation frames are 640×512 crops of the 1280×720 feed. We reproduce that transform on training frames (crop $x \in [192, 1088]$, resize to 640×512) so pseudo-labelling, training and inference share one geometry.

Weak labels. We parse `tools.csv` per robot arm into the multiset of tools installed at every second, resolving overlaps on one arm in favour of the shorter interval.

Pseudo-labelling. A teacher ensemble of two RT-DETR-L [1] models predicts on every extracted frame; we fuse the predictions with weighted box fusion (WBF, IoU 0.7, `conf_type` max) [2]. We keep boxes with confidence ≥ 0.20 whose class appears in the installed multiset, drop boxes centred in the da Vinci interface banners, and drop whole frames in which two different classes share a box (IoU > 0.7). A frame is accepted only if the per-class box count equals the installed count exactly and every kept box has confidence ≥ 0.35 . We apply *no* box-size filter: an earlier variant clipping box areas to the 2nd–98th percentile removed the near- and far-field scale samples the detector needs and reduced AP75.

Iteration. Retraining the teacher on its own output raised the acceptance rate from 26% to 41% and the pool from 82k to 142k frames. Iterative pseudo-labelling converges here because `tools.csv` is an external, model-independent filter.

Background-collapse failure. Prograsp forceps, stapler and tip-up grasper were initially dropped from the pseudo-label targets while their images were kept, teaching the detector to treat visible instances as background: 65,182 of 149,569 frames listed one of them as installed but carried no box, against 711 positive prograsp boxes. Deleting those frames raised held-out detection of prograsp and stapler from 0.000 to 0.255 and 0.265.

Inference. The submitted container runs the 640 and 800 models and fuses their per-frame detections with WBF, emitting at most 100 boxes per frame at confidence ≥ 0.05 with true confidence values. A wall-clock guard falls back to the single 640 model if the projected runtime exceeds the limit.

Category 2. We classify each question into one of four routes from the position of its auxiliary verb and wh-word, then emit a templated sentence: BERTScore against short reference answers rewards matching the expected answer *form*. Yes/no polarity for five high-recall tools is decided visually by our C1 detector over 15 sampled frames; the rest fall back to a prior estimated from 30-second windows of `tools.csv`.

0.1.2 Model Training

All detectors are RT-DETR-L (Ultralytics 8.4.137) initialised from COCO-pretrained weights and trained on one NVIDIA A100 SXM4 40GB. We set the optimizer explicitly to AdamW ($\text{lr}_0 = 5.56 \times 10^{-4}$, $\text{lr}_f = 0.01$, momentum 0.9, weight decay 5×10^{-4} , `warmup_bias_lr` = 0): Ultralytics silently switches `optimizer=auto` to MuSGD at lr 0.01 above 10,000 iterations, which would have changed the recipe as the dataset grew. The submitted models use 16 epochs, `close_mosaic` 6, batch 24 at 640 (5.8h) and batch 20 at 800 (7.6h), with default mosaic, scale and flip augmentation. The final training set is 93,660 frames. For ablations we held official validation videos 1 and 7 out of the student; for the submitted models we merged all annotated official videos into training and submit `last.pt`.

0.1.3 Preliminary Performance

Table 1 lists our official scores; data scale-up produced the largest single gain. On the Final set, routing the four never-detected classes to a separate model was worth +0.0003, while 800-pixel training with cross-model fusion was worth +0.0100.

In paired ablations on a student hold-out (official videos 1 and 7), 800-pixel training gave +0.0131 mean mAP and +0.0275 AP75, and cross-model WBF gave +0.0166 and +0.0376 on a pair of 640-pixel models, not on the 640/800 pair we submitted; both gains sit at high IoU, i.e. they tighten boxes rather than find more tools. What did not work: 39 inference-time variants had a ceiling of +0.005; ten temporal post-processing variants were all negative; fine-tuning on official ground truth alone was worse than mixed training.

Limitations. These ablations isolate the student, not the pipeline: the teacher had seen the held-out videos, so absolute values are optimistic. We also saw a proxy metric invert: tool-presence set-F1 on 2,820 unseen frames rose from 0.578 to 0.819 for a relabelling scheme that was -0.0127 on official-protocol mAP. Session-level splits and a small human-annotated box set for the rare classes are the next steps.

Disclosure. No external datasets were used. Bounding boxes for the three never-detected classes were annotated with a large language model (Claude Opus 5): 718 frames annotated, 287 accepted, expanded to

Phase	Configuration	mAP
C1 Preliminary	official validation videos only	0.4372
C1 Preliminary	domain-aligned strict pseudo-labels	0.4464
C1 Preliminary	pseudo-label source 20 $\rightarrow$ 280 videos	0.5059
C1 Final	single model, 640	0.4830
C1 Final	+ separate model for four zero-AP classes	0.4833
C1 Final	+ WBF of the 640 and 800 models	0.4930
C2 Final	echo templates	0.5833
C2 Final	+ question-type routing fix	0.6067
C2 Final	+ visual yes/no polarity	0.6191

Table 1: Official scores. C1 is mAP@[.5:.95] averaged over videos; C2 is BERTScore-F1.

2,096 training frames by template matching. A further 202 grasping-retractor frames came from 32 hand-seeded boxes propagated the same way, giving 2,298 model-assisted frames in total (2.5% of the training set). The annotation protocol and code are in our repository. A variant that read the da Vinci interface with EasyOCR to gate pseudo-labels at training time was evaluated but is not part of the submitted model.

References

- [1] Yian Zhao, Wenyu Lv, Shangliang Xu, Jinman Wei, Guanzhong Wang, Qingqing Dang, Yi Liu, and Jie Chen. DETRs beat YOLOs on real-time object detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16965–16974, 2024.
- [2] Roman Solovyev, Weimin Wang, and Tatiana Gabruseva. Weighted boxes fusion: Ensembling boxes from different object detection models. *Image and Vision Computing*, 107:104117, 2021.